

NEHA GUPTA

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EDUCATION

University of Georgia	Athens, GA
<i>Doctor of Philosophy (PhD in Bioinformatics)</i>	GPA: 4.00/4.00 Aug 2023 - Present
Boston University	Boston, MA
<i>Master of Science in Bioinformatics</i>	GPA: 3.65/4.00 Sep 2020 - Jan 2022
Amity University	Noida, India
<i>Bachelor of Technology in Biotechnology</i>	GPA: 7.61/10.00 Jul 2015 - Jun 2019

TECHNICAL SKILLS

Bioinformatics Techniques: Single cell RNA-seq (scRNA), Single nucleus RNA-seq (snRNA), Bulk RNA-seq, Drug discovery & Molecular docking, Protein modeling

Programming & Scripting: R, Python, Java, Shell scripting

R & Bioconductor Packages: Seurat, DESeq2, dplyr, tidyverse, ggplot2, enrichR, stringr, pubmed.mineR, Biomart, Harmony

Python modules:

Computing: Cluster Computing (AWS), Linux/UNIX, Windows

RESEARCH EXPERIENCE

<i>Graduate Research Assistant</i>	University of Georgia, Athens, GA	Aug 2023 – Present
Advisor: Dr. Natarajan Kannan		
- Conducting single-cell isoform and transcriptomic analysis to elucidate kinase function in neuronal cells affecting neurodegenerative diseases - Performing computational protein structural analysis using PyMOL and AlphaFold3 to characterize kinase functional motifs and catalytic activity		
<i>Research Associate II, Bioinformatics</i>	Cellarity Inc., Somerville, MA	Feb 2022 - Jun 2023
Advisor: Dr. Scott Steelman		
- Analyzed transcriptome, single-cell, and Illumina sequencing data supporting laboratory operations and workflow prototyping - Designed and implemented data QC analysis modules and managed bioinformatics data processing pipelines		
<i>Computational Biology Intern</i>	Broad Institute of MIT & Harvard, Cambridge, MA	Jun 2021 - Dec 2021
Advisor: Dr. Ayshwarya Subramanian		
- Executed computational analysis on high-throughput sequencing datasets (scRNA-seq, nuc-seq, Bulk RNA, ChIP-seq) from in-vitro and in-vivo disease models - Performed preprocessing, statistical analysis, functional annotation, and visualization of multi-project datasets focusing on TIM-3 gene		
<i>Bioinformatics Analyst</i>	Eminent Biosciences, Indore, India	Oct 2019 - Aug 2020
Advisor: Anuraj Nayarisseri		
Implemented transcriptome expression analysis pipeline using HISAT2 and StringTie for 9 paired normal and colon cancer tissue samples, performing differential gene expression analysis with DESeq2		

PUBLICATIONS

- Lu, J. D., Zhao, P., Prasad, A., **Gupta, N.**, Gravel, N., Shidhaye, T., ... & Kannan, N. (2025). Deciphering the unique autoregulatory mechanisms and substrate specificity of the understudied DCLK3 kinase linked to neurodegenerative diseases. *Journal of Biological Chemistry*, 301(10).
- Kimura, K., Subramanian, A., Yin, Z., **Gupta, N.**, Wu, Y., He, D., ... & Kuchroo, V. K. (2025). Immune checkpoint TIM-3 regulates microglia and Alzheimer's disease. *Nature*, 641(8063), 718-731
- Gupta N** †, Choudhary SK †, Bhagat N, Karthikeyan M, Chaturvedi A. In Silico Prediction, Molecular Docking and Dynamics Studies of Steroidal Alkaloids of *Holarrhena pubescens* Wall. ex G. Don to Guanylyl Cyclase C: Implications in Designing of Novel Antidiarrheal Therapeutic Strategies. *Molecules*. 2021;26(14):4147

AWARDS & CONFERENCES

- Served as President of Indian Student Association (ISA) at UGA for 2024-2025
- Received 'Student Leader of the Year 2025' by International Student Life (ISL) at UGA.
- Poster Presentation at American Society of Human Genetics & Genomics (ASHG) in 2025 in Boston, MA
- Mentored NSF REU student for research project in summer 2025.
- Awarded Beverly B. Ledakowski Cancer Research Fellowship in 2026.